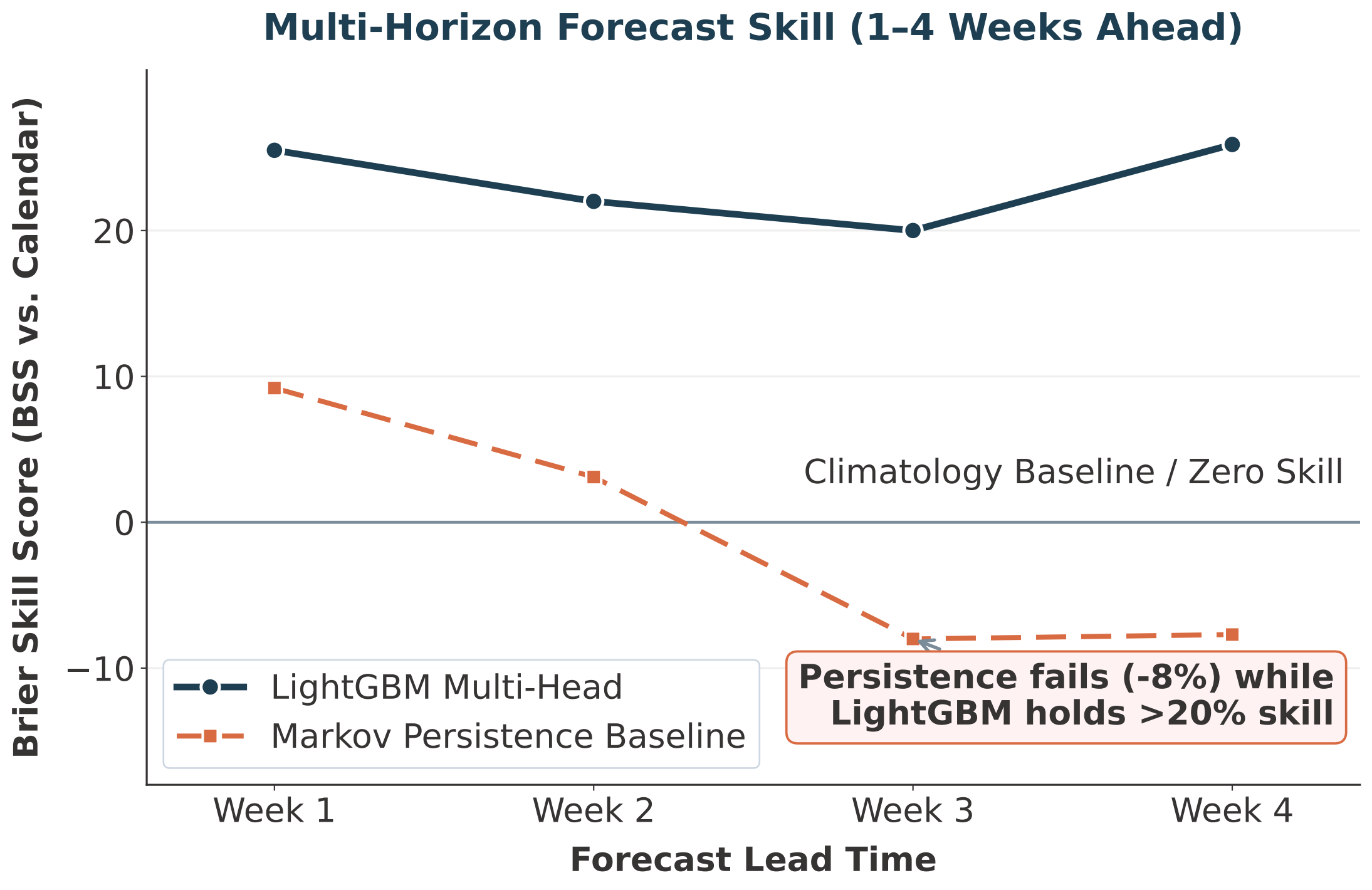
Multi-Horizon Forecast Skill (1–4 Weeks Ahead)
Brier Skill Score (BSS vs. Calendar)
20
10
0
−10
Climatology Baseline / Zero Skill
LightGBM Multi-Head
Markov Persistence Baseline
Persistence fails (-8%) while
LightGBM holds >20% skill
Week 1
Week 2
Week 3
Week 4
Forecast Lead Time